

# WASP-36b

R-1955

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-36\_151204 · created 2026-08-06T15:57:52+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457359.3311 +/- 0.0017 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.165 +/- 0.049                 |
| Transit depth (Rp/Rs)^2                           | 2.71 +/- 1.61 [%]               |
| Orbital Inclination (inc)                         | 87.75 +/- 2.91                  |
| Airmass coefficient 1 (a1)                        | 7.51 +/- 4.35                   |
| Airmass coefficient 2 (a2)                        | -0.87 +/- 0.36                  |
| Scatter in the residuals of the lightcurve fit is | 45.18 %                         |
| Best Comparison Star                              | #3 - [396, 478]                 |
| Optimal Aperture                                  | 2.88                            |
| Optimal Annulus                                   | 5.75                            |
| Transit Duration (day)                            | 0.0189 +/- 0.0045               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.165 ± 0.049 vs 0.1368 (deviation 0.56σ) |
| Duration fit vs published | 0.0189 d vs 0.0758 d (deviation 12.32σ)   |
| Mid-time O–C              | -2214.3 min                               |
| Depth SNR                 | 3.3                                       |
| Residual scatter          | 45.18 %                                   |

### Light curve

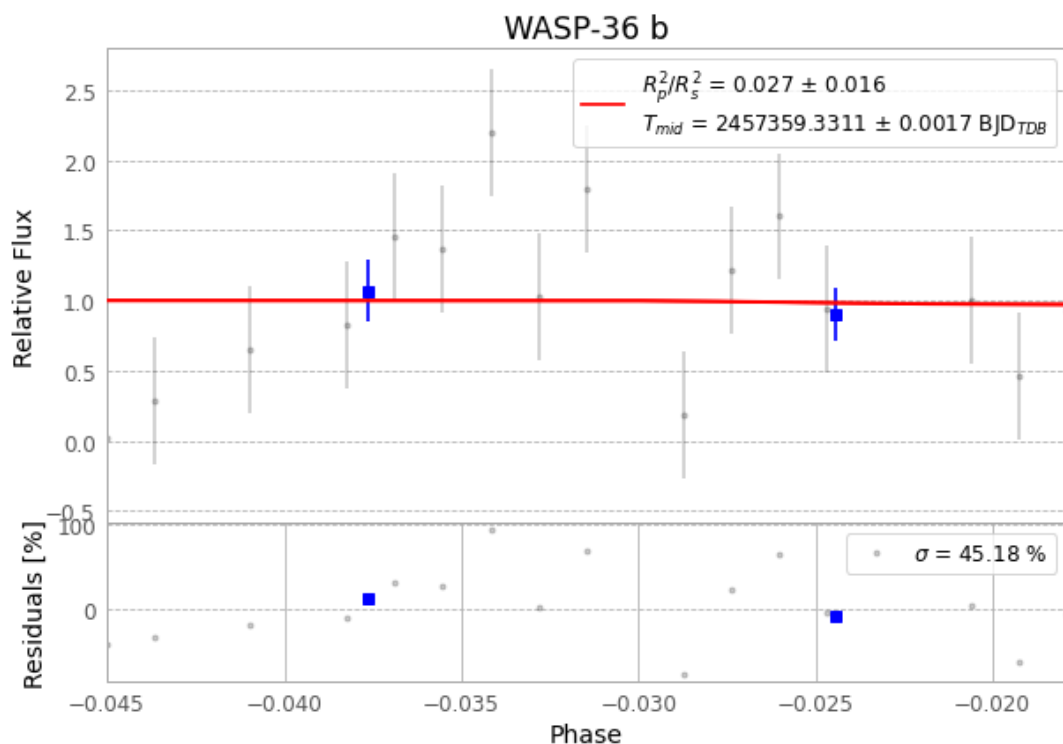


Figure 1 — FinalLightCurve\_WASP-36 b\_2015-12-03.png (SHA-256 3a9b6b1a84257163...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 353], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 eb4c2b1b2e463b55...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit fd74a91

## Data provenance

77 FITS frames (51 MB), WASP-36151204065712.FITS → WASP-36151204104512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-151204.log (9dfd7bee7571...), FinalLightCurve\_WASP-36 b\_2015-12-03.png (3a9b6b1a8425...), FinalLightCurve\_WASP-36 b\_2015-12-03.csv (bac7c3059ab6...)

## Compute resources

Wall time 107.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-06 15:57Z.